



ŘEDITELSTVÍ SILNIC A DÁLNIC ČR

**NDIC - DATEX II Elaborated Data
Publication - FCD**

Release 1.1rev2

National Traffic Information Centre

2023-03-09

CONTENTS

1	Introduction	2
1.1	General terms	2
2	Concepts	4
2.1	Publications of type ElaboratedDataPublication	4
2.2	Calculated FCD information	5
2.3	Providing only currently available data	10
3	Temporal validity	11
3.1	FCD input data age	11
4	Location Description	13
5	Annex A: Examples of messages	14
5.1	samples/fcd-all-lorry-car-sample.xml - FCD data on 1 segment	14
5.2	samples/fcd-all-lorry-car-sample-no-data.xml - no FCD data on 1 segment	21
6	Appendix B: W3C XML Schema	24
6.1	<i>schema.xsd</i>	24
7	Changelog	36
	Bibliography	37
	Index	38

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cz-ndic_d2-fcd-v1.1

Format

1.1

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1.1rev2

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DATEX II v2.3/ElaboratedDataPublication

Extends

elaboratedData, basicData/type=TrafficStatus, basicData/type=TraveTimeData

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Publisher

National Traffic Information Centre

Author

Jan Vlčinský, Petr Bureš

This specification defines format, used for publishing of FCD generated data, e.g. travel times, speed, traffic statuses and quality parameters on a set of predefined locations, separately for different types of vehicles (cars and trucks).

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format

description of data structure for transferring some kind of information

protocol

description of data transfer process and rules

NDIC

national traffic information centre

DATEX II model

class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [\[d2ref\]](#). For additional information see [\[d2supp\]](#).

(DATEX II) profile

simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension

extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension

Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [16157-1]

C level extension

Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication

an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [16157-3], *ElaboratedDataPublication* [16157-5], *PredefinedLocationsPublication* [16157-3] etc.

(W3C) XML schema

language, which allows definition of rules for a XML document structure.

XSD

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CONCEPTS

The described format specifies provision of floating car data (FCD) information. This includes travel times, traffic flow speeds, traffic levels and estimation of congestion lengths on a set of predefined locations.

Unlike the previous version of the format *x-format:cz-ndic_d2-fcd-v1.0*, it introduces **distinction by vehicle type (all, car, lorry)**.

2.1 Publications of type *ElaboratedDataPublication*

The format is based on the DATEX II publication type *ElaboratedDataPublication* [16157-5] and uses it to describe information about **travel times, speeds and traffic levels** on a set of predefined locations, differentiated by vehicle type.

The information about **traffic levels, length of queues** with their **start and end** is added via the level B extension of the *TrafficStatus* class of the DATEX II model.

Information about **data quality** and **number of FCD vehicles** on the segment, including their type, is added via the level B extension of the *TravelTimeData* class of the DATEX II model.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsi=
   ↳ "http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://datex2.eu/
   ↳ schema/2/2_0 ../schema/schema.xsd">
3   <exchange>
4     <supplierIdentification>
5       <country>cz</country>
6       <nationalIdentifier>ŘSD</nationalIdentifier>
7     </supplierIdentification>
8   </exchange>
9   <payloadPublication lang="cs" xsi:type="ElaboratedDataPublication">
10    <feedType>x-format:cz-ndic_d2-fcd-v1.1</feedType>
11    <publicationTime>2022-12-18T00:00:06+01:00</publicationTime>
12    <publicationCreator>
13      <country>cz</country>
14      <nationalIdentifier>ŘSD</nationalIdentifier>
15    </publicationCreator>
16    <periodDefault>300</periodDefault>
17    <timeDefault>2022-12-17T23:59:35+01:00</timeDefault>
18    <headerInformation>
19      <confidentiality>noRestriction</confidentiality>
20      <informationStatus>real</informationStatus>
21    </headerInformation>
```

FCD information for a particular segment is listed in 3 *elaboratedData* elements, (they can only hold **one type of information at one time**):

- traffic density in the *TrafficStatus* class + traffic levels (by CZ) and congestions in its extension,

- traffic speed in class *TrafficSpeed* and
- travel time in class *TravelTimeData* + quality parameters in its extension.

Note: In the case of segments where status information is calculated **separately by vehicle type**, the number of *elaboratedData* elements is **tripled**, i.e. a total of 9 elements (3x *anyVehicle*, 3x *car* and 3x *lorry*) per segment at one time.

The *elaboratedData* element does not have its own unique identifier and is uniquely determined by the publication time and the referenced location identifier.

The identification of the publication profile is given in the *feedType* element referring to the specific documentation published in the NAP (national access point).

2.2 Calculated FCD information

Status information is given for all vehicles together (*anyVehicle*) and separately (*car*, *lorry*) according to the type of location:

- Segments with **mixed traffic flow**, where it does not make sense to evaluate traffic parameters separately by vehicle type, i.e. vehicles moves in a mixed traffic or in single lane traffic. For these segments only the elements with *anyVehicle* are present.
- Segments with **separate traffic flow** where it makes sense to evaluate traffic parameters separately by vehicle type, i.e. vehicles move in separated traffic streams, in multiple lanes, so separated queues of vehicles by vehicle type may occur. For these segments, the all elements with *anyVehicle* are present and all elements with *car* and *lorry* are provided depending on **whether the vehicles of the type were present on the segment at the time of evaluation**.

The status information included in this profile is:

- traffic flow speed [km/h] per
 - actual conditions
 - in free flow conditions (freeFlow)
- travel times [s] per
 - actual conditions
 - in free flow conditions (freeFlow)
- actual delay [s]
- traffic density [-]

Other information included in this profile is:

- system response times [s]

The status information included in this profile by extension of the *TrafficStatus* class is:

- traffic levels [-] (1-5 as used in the Czech Republic)
- queues on the segment, estimation of their length [m] and estimation of the start [%] and end [%] of the queue on the referenced segment

The common qualitative parameters included in this profile by extending the *TravelTimeData* class are:

- data reliability level [-]
- number of FCD vehicle probes on the segment [-]

Information included by the extension is shown in Figure 1.

The following articles are dedicated to the detailed description of the format.

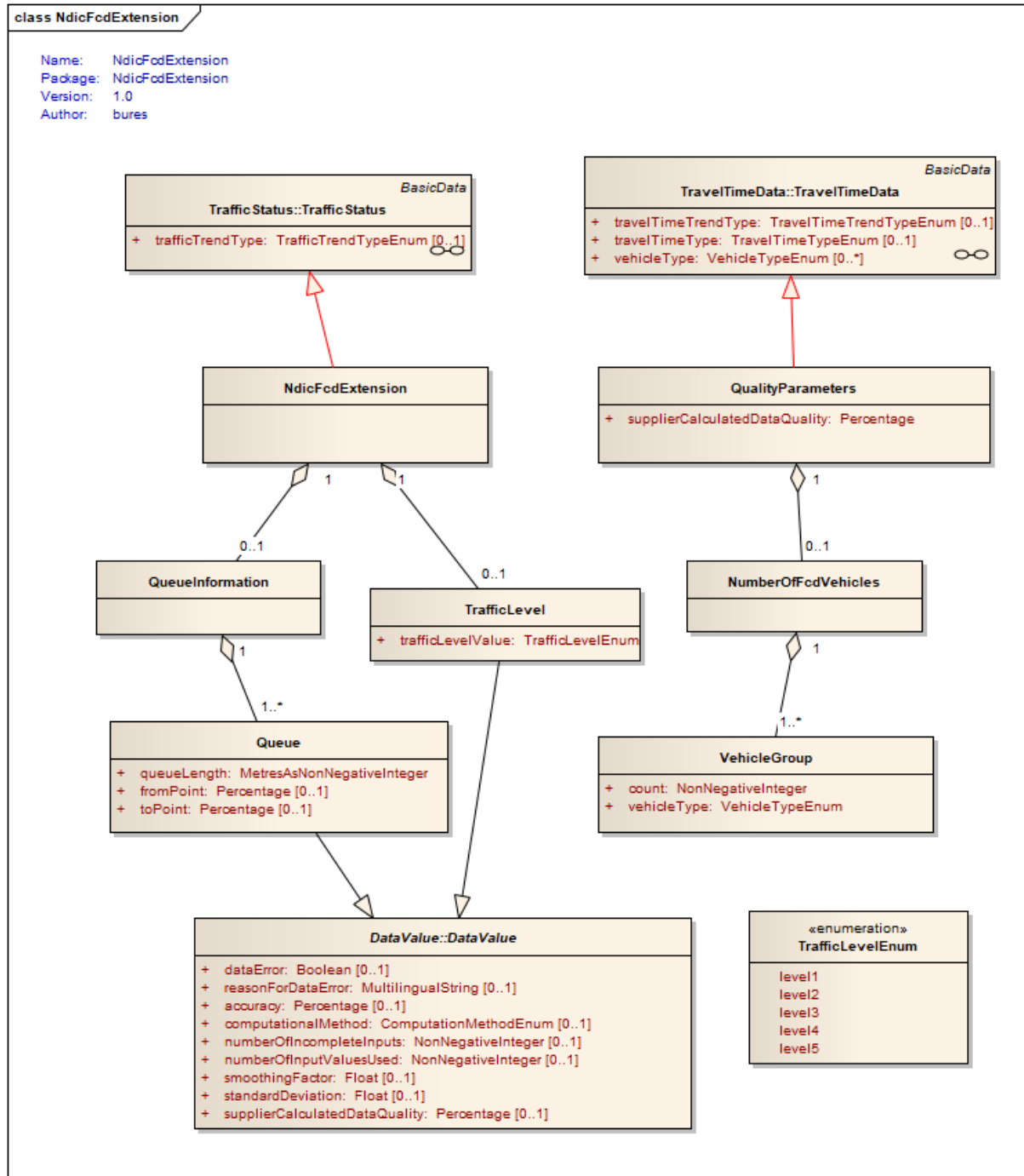


Fig. 1: TrafficStatus and TravelTimeData class extension.

2.2.1 Vehicle category

Traffic parameters are calculated separately for passenger (*car*) and truck (*lorry*) and all vehicles (*anyVehicle*) in separate *elaboratedData* elements, identified by *vehicleType* value.

The *TrafficStatus* class element had to be extended by the *vehicleType* item in the type B extension.

```
<ndicFcdExtension>
  <vehicleType>anyVehicle</vehicleType>
```

The *TrafficSpeed* class element uses *forVehiclesWithCharacteristicsOf* including *vehicleType*.

```
<forVehiclesWithCharacteristicsOf>
  <vehicleType>anyVehicle</vehicleType>
</forVehiclesWithCharacteristicsOf>
```

The *TravelTimeData* class element directly contains the *vehicleType* element.

```
<vehicleType>anyVehicle</vehicleType>
<!-- travel time information -->
```

2.2.2 Data Reliability Measure

The **Data Reliability Measure** characterizes the quality of the data sample and has values <0-1> where 1 means 100% reliable data. The quality of the data sample is evaluated by the age and number of the input values. The calculation formula is not part of this profile.

The data reliability rate is given for each calculated value separately by the *supplierCalculatedDataQuality* attribute.

```
<freeFlowSpeed numberOfInputValuesUsed="9999" supplierCalculatedDataQuality=
  "0.8">
```

Note: In this profile the extension for the Data Reliability Measure is not used.

2.2.3 Number of vehicle probes

The actual **number of vehicles probes** is given for each calculated value separately by its *numberOfInputValuesUsed* attribute.

```
<trafficStatus numberOfInputValuesUsed="9" supplierCalculatedDataQuality="1">
  <trafficStatusValue>congested</trafficStatusValue>
</trafficStatus>
```

Some statistics, such as *freeFlowSpeed* or *freeFlowTravelTime*, are independent of the number of vehicle probes. In this case, the value of the number of vehicles, if specified by the *numberOfInputValuesUsed* attribute, is set to 9999.

```
<freeFlowTravelTime numberOfInputValuesUsed="9999"
  supplierCalculatedDataQuality="1">
  <duration>259</duration>
</freeFlowTravelTime>
```

2.2.4 Traffic flow speed

The **traffic flow speed** [km/h] on the segment is given by two separate *elaboratedData* elements.

The actual average speed *averageVehicleSpeed* in the measured time interval is given in the *TrafficSpeed* class of the *basicData* element.

```

123     <averageVehicleSpeed numberOfInputValuesUsed="9"
124     ↪supplierCalculatedDataQuality="1">
125         <speed>38</speed>
        </averageVehicleSpeed>

```

The *freeFlowSpeed* [km/h] on the segment is specified in the *TravelTimeData* class in the *basicData* element.

```

176     <freeFlowSpeed numberOfInputValuesUsed="9999" supplierCalculatedDataQuality=
177     ↪"0.8">
178         <speed>50</speed>
        </freeFlowSpeed>

```

2.2.5 Travel time and actual delay

The **Travel Time** [s] on the segment is given by one *elaboratedData* element.

Current average travel time *travelTime*.

```

168     <travelTime numberOfInputValuesUsed="9" supplierCalculatedDataQuality="1">
169         <duration>549</duration>
170     </travelTime>

```

Average free flow travel time *freeFlowTravelTime*.

```

172     <freeFlowTravelTime numberOfInputValuesUsed="9999"
173     ↪supplierCalculatedDataQuality="1">
174         <duration>259</duration>
        </freeFlowTravelTime>

```

The **actual delay** is expressed implicitly as the difference between the values of the actual average travel time (*travelTime*) and the free flow travel time (*freeFlowTravelTime*).

2.2.6 Traffic levels

Traffic status (level of service) is expressed by *TrafficStatusEnum* enumeration as defined in standard [16157-3] in table A.125.

```

34     <trafficStatus numberOfInputValuesUsed="9" supplierCalculatedDataQuality="1">
35         <trafficStatusValue>congested</trafficStatusValue>
36     </trafficStatus>

```

Table 1: Values of the “TrafficStatusEnum” enumeration.

Value	Meaning	Definition
impossible	Traffic collapse	Vehicles on the roads are stationary or in convoys only moving very slowly, the average speed is very low.
congested	Traffic congestion	Vehicle queues are formed, traffic is not flowing smoothly, average speed is significantly reduced, intersection passage is impaired.
heavy	Heavy traffic	Streams of vehicles form, traffic is flowing, but speed is below the maximum speed limit.
freeFlow	Flowing traffic	Traffic is flowing in that direction. Traffic is moving at or above the speed limit and drivers are free to change lanes.
unknown	Unknown	Traffic conditions are unknown.

The *TrafficStatus* class is extended (level B extension) to explicitly specify the traffic levels as defined and used in the Czech Republic. The traffic levels (1-5) are specified in the *TrafficLevel* class using the phrases “level1” to “level5”, in the *trafficLevelValue* element.

```

<vehicleType>anyVehicle</vehicleType>
<trafficLevel>
  <trafficLevelValue>level4</trafficLevelValue>
</trafficLevel>

```

The *TrafficLevel* is calculated as the % difference between the current traffic flow speed and the traffic flow speed on the free segment.

Table 2: Calculation in the definition of traffic stages in the FCD publication

Degree	Description	% of difference
1	single vehicle operation only, driving is smooth	<85 %, 100 %)
2	small groups of vehicles, smooth running, passing intersections without problems	<75 %, 85 %)
3	vehicle streams are forming, traffic is smooth but speed is below the maximum speed limit	<50 %, 75 %)
4	vehicle queues are forming, traffic is not flowing, average speed is significantly reduced, passing the intersections is impaired	<20 %, 50 %)
5	traffic collapse - vehicles are stationary on the roads or moving very slowly in queues, average speed is very low	<0 %, 20 %)

2.2.7 Traffic queue

The **queues** on a segment are expressed by the presence of the *queueInformation* element, which was added to the *TrafficStatus* class via the DATEX II type B extension.

Segments are internally divided into smaller parts (sections of the Global Network, GN road network). The presence of congestion is evaluated with respect to GN sections as a decrease in the travelled speed of this section **below 30 % of the free-flow speed**.

The *queueInformation* element contains aggregated values from individual congested sections of the segment. If none of the sections of the segment were congested, the *queueInformation* element is not present. The element is present only once per segment, the values of its attributes are adapted to this (for more, see the description of the attributes and the image below).

```

<queueInformation>
  <queue>
    <queueLength>148</queueLength>

```

(continues on next page)

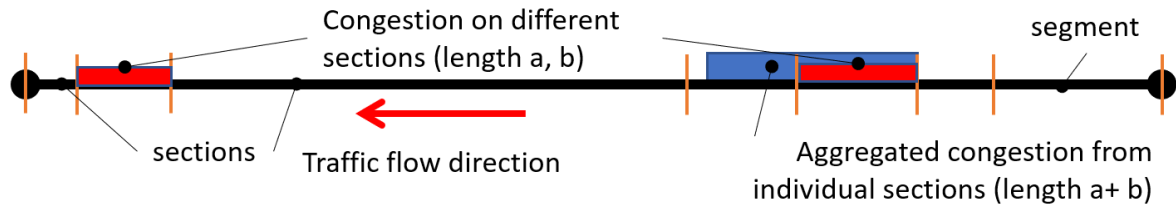


Fig. 2: Representation of the aggregation of individual congestions on a segment

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```

47      <!-- start of the queue (tail) at the 0.4 of the length of the segment -->
48      <segment -->
49      <fromPoint>0.4</fromPoint>
50      <!-- end of the queue (head) at the 0.9 of the length of the segment -->
51      <toPoint>0.9</toPoint>
52      </queue>
      </queueInformation>

```

The length of a queue on a referenced segment is expressed in the *queueLength* element and is aggregated length of the traffic congestions for all sections of the segment [m]. i.e. the sum of the lengths of the **congested** Global Network sections (even non-contiguous) of the segment.

The start and end of the queue is given by the optional *fromPoint* and *toPoint* attributes, expressing relative distance from the beginning of the segment in the direction of traffic. A queue, according to this definition, starts (*fromPoint*) at the point where vehicles enter it and ends (*toPoint*) at the point where they leave it and continue on their route. See Figure 2.

In the case of multiple split congestions on a segment, for example, a congestion detected on GN sections 1 and 3, the *fromPoint* attribute only gives information about the 1st congestion and *toPoint* is calculated by using the *queueLength* value.

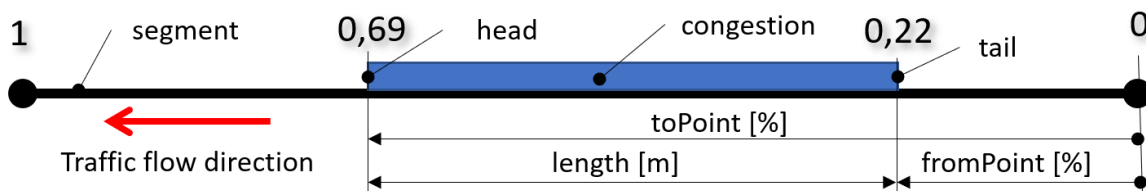


Fig. 3: Example of referencing the relative position of a queue on a segment.

2.3 Providing only currently available data

In case of unavailability of input data on a given segment for the status information calculation, only free flow traffic parameters are transmitted, in the *basicData* element of the *TravelTimeData* class.

Thus, a segment for which actual status data exists is covered by 9 *elaboratedData* elements, while a segment for which no actual data exists is covered by 3 elements.

Note: 9 *elaboratedData* elements for one segment is a maximum number. The number could be lower (6) if for example only cars or only lorries are present on the segment.

TEMPORAL VALIDITY

The publication is periodically updated, the value in the *periodDefault* element specifies the duration of the update interval in seconds.

The value of *publicationTime* specifies when the publication was created, but **not the reference time of the calculation**.

The *timeDefault* value specifies the common reference **calculation time** for all record values and is the end time of a particular measurement and calculation period.

No other determination of data validity is used.

```
11 <publicationTime>2022-12-18T00:00:06+01:00</publicationTime>
12 <publicationCreator>
13   <country>cz</country>
14   <nationalIdentifier>ŘSD</nationalIdentifier>
15 </publicationCreator>
16 <periodDefault>300</periodDefault>
17 <timeDefault>2022-12-17T23:59:35+01:00</timeDefault>
```

3.1 FCD input data age

Each traffic parameter indicates the time of the most recent input information used for its calculation in the *measurementOrCalculationTime* element.

Note: The input data for the calculation of traffic parameters are the positions of the vehicle probes. These data are evaluated at regular time intervals, recalculated into sections travelled (parts of the segment) and values for such sections travelled. The last update of the segment status occurs when the last vehicle leaves the segment.

The age of the input data used for each record can be calculated as the difference between the *timeDefault* of the entire publication and the *measurementOrCalculationTime* value of that record. The age of the input data can be used to determine **system response time**.

In case of insufficient underlying data for the calculation of the statuses (according to the calculation algorithm), elements containing information on **actual** speed, traffic statuses and travel times are omitted.

```
16 <periodDefault>300</periodDefault>
17 <timeDefault>2022-12-17T23:59:35+01:00</timeDefault>
18 <headerInformation>
19   <confidentiality>noRestriction</confidentiality>
20   <informationStatus>real</informationStatus>
21 </headerInformation>
22 <!-- information for one segment -->
23 <!-- traffic status element - ALL vehicles -->
24 <elaboratedData>
```

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```
25 <source>
26   <sourceIdentification>ndic-secid-1-01</sourceIdentification>
27 </source>
28 <basicData xsi:type="TrafficStatus">
29   <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
  measurementOrCalculationTime>
```

LOCATION DESCRIPTION

FCD data are measured at predefined road segments. The segment can be given by 2 consecutive ALERT-C location points, sections of the road network used (GlobalNetwork), or requirements based on the physical layout of the road.

The location of FCD segment is expressed by a reference in a *pertinentLocation* element to a set of predefined locations in a *PredefinedLocationsPublication* [16157-3] associated with this information source.

```
30 <pertinentLocation xsi:type="LocationByReference">
31   <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
    ↪targetClass="PredefinedLocation" version="0"/>
32 </pertinentLocation>
```

In addition to describing the location with the *PredefinedLocation* element, the format optionally allows the location to be described with an additional reference, the provider's internal identifier, in the *source* element.

```
25 <source>
26   <sourceIdentification>ndic-secid-1-01</sourceIdentification>
27 </source>
```

Location description methods are beyond the scope of this document and are described elsewhere.

ANNEX A: EXAMPLES OF MESSAGES

5.1 samples/fcd-all-lorry-car-sample.xml - FCD data on 1 segment

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsi=
  ↪ "http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://datex2.eu/
  ↪ schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="ElaboratedDataPublication">
    <feedType>x-format:cz-ndic_d2-fcd-v1.1</feedType>
    <publicationTime>2022-12-18T00:00:06+01:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </publicationCreator>
    <periodDefault>300</periodDefault>
    <timeDefault>2022-12-17T23:59:35+01:00</timeDefault>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <!-- information for one segment -->
    <!-- traffic status element - ALL vehicles -->
    <elaboratedData>
      <source>
        <sourceIdentification>ndic-secid-1-01</sourceIdentification>
      </source>
      <basicData xsi:type="TrafficStatus">
        <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
  ↪ measurementOrCalculationTime>
        <pertinentLocation xsi:type="LocationByReference">
          <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
  ↪ targetClass="PredefinedLocation" version="0"/>
        </pertinentLocation>
        <!-- traffic status information -->
        <trafficStatus numberOfInputValuesUsed="9" supplierCalculatedDataQuality="1">
          <trafficStatusValue>congested</trafficStatusValue>
        </trafficStatus>
        <!-- extension: traffic level and queue information -->
```

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```

<trafficStatusExtension>
  <ndicFcdExtension>
    <vehicleType>anyVehicle</vehicleType>
    <trafficLevel>
      <trafficLevelValue>level4</trafficLevelValue>
    </trafficLevel>
    <queueInformation>
      <queue>
        <queueLength>148</queueLength>
        <!-- start of the queue (tail) at the 0.4 of the length of the
->segment -->
        <fromPoint>0.4</fromPoint>
        <!-- end of the queue (head) at the 0.9 of the length of the segment -
->->
        <toPoint>0.9</toPoint>
      </queue>
    </queueInformation>
  </ndicFcdExtension>
</trafficStatusExtension>
</basicData>
</elaboratedData>
<!-- traffic status element - CARS -->
<elaboratedData>
  <basicData xsi:type="TrafficStatus">
    <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
->measurementOrCalculationTime>
    <pertinentLocation xsi:type="LocationByReference">
      <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
->targetClass="PredefinedLocation" version="0"/>
    </pertinentLocation>
    <trafficStatus numberOfInputValuesUsed="5" supplierCalculatedDataQuality="0.
->850">
      <trafficStatusValue>congested</trafficStatusValue>
    </trafficStatus>
    <trafficStatusExtension>
      <ndicFcdExtension>
        <vehicleType>car</vehicleType>
        <trafficLevel>
          <trafficLevelValue>level4</trafficLevelValue>
        </trafficLevel>
        <queueInformation>
          <queue>
            <queueLength>89</queueLength>
            <fromPoint>0.6</fromPoint>
            <toPoint>0.9</toPoint>
          </queue>
        </queueInformation>
      </ndicFcdExtension>
    </trafficStatusExtension>
  </basicData>
</elaboratedData>
<!-- traffic status element - LORRIES -->
<elaboratedData>
  <basicData xsi:type="TrafficStatus">
    <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
->measurementOrCalculationTime>

```

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```

    <pertinentLocation xsi:type="LocationByReference">
      <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
↳targetClass="PredefinedLocation" version="0"/>
    </pertinentLocation>
    <trafficStatus numberOfInputValuesUsed="4" supplierCalculatedDataQuality="0.
↳750">
      <trafficStatusValue>congested</trafficStatusValue>
    </trafficStatus>
    <trafficStatusExtension>
      <ndicFcdExtension>
        <vehicleType>lorry</vehicleType>
        <trafficLevel>
          <trafficLevelValue>level4</trafficLevelValue>
        </trafficLevel>
        <queueInformation>
          <queue>
            <queueLength>148</queueLength>
            <fromPoint>0.4</fromPoint>
            <toPoint>0.9</toPoint>
          </queue>
        </queueInformation>
      </ndicFcdExtension>
    </trafficStatusExtension>
  </basicData>
</elaboratedData>
<!-- speed information element - ALL vehicles -->
<elaboratedData>
  <basicData xsi:type="TrafficSpeed">
    <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
↳measurementOrCalculationTime>
    <pertinentLocation xsi:type="LocationByReference">
      <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
↳targetClass="PredefinedLocation" version="0"/>
    </pertinentLocation>
    <!-- vehicle category -->
    <forVehiclesWithCharacteristicsOf>
      <vehicleType>anyVehicle</vehicleType>
    </forVehiclesWithCharacteristicsOf>
    <!-- average vehicle speed information -->
    <averageVehicleSpeed numberOfInputValuesUsed="9"
↳supplierCalculatedDataQuality="1">
      <speed>38</speed>
    </averageVehicleSpeed>
  </basicData>
</elaboratedData>
<!-- speed information element - CARS -->
<elaboratedData>
  <basicData xsi:type="TrafficSpeed">
    <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
↳measurementOrCalculationTime>
    <pertinentLocation xsi:type="LocationByReference">
      <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
↳targetClass="PredefinedLocation" version="0"/>
    </pertinentLocation>
    <forVehiclesWithCharacteristicsOf>
      <vehicleType>car</vehicleType>

```

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```

        </forVehiclesWithCharacteristicsOf>
        <averageVehicleSpeed numberOfInputValuesUsed="5"
↳supplierCalculatedDataQuality="0.850">
            <speed>58</speed>
        </averageVehicleSpeed>
    </basicData>
</elaboratedData>
<!-- speed information element - LORRIES -->
<elaboratedData>
    <basicData xsi:type="TrafficSpeed">
        <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
↳measurementOrCalculationTime>
        <pertinentLocation xsi:type="LocationByReference">
            <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
↳targetClass="PredefinedLocation" version="0"/>
        </pertinentLocation>
        <forVehiclesWithCharacteristicsOf>
            <vehicleType>lorry</vehicleType>
        </forVehiclesWithCharacteristicsOf>
        <averageVehicleSpeed numberOfInputValuesUsed="4"
↳supplierCalculatedDataQuality="0.750">
            <speed>18</speed>
        </averageVehicleSpeed>
    </basicData>
</elaboratedData>
<!-- travel time element - ALL vehicles -->
<elaboratedData>
    <basicData xsi:type="TravelTimeData">
        <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
↳measurementOrCalculationTime>
        <pertinentLocation xsi:type="LocationByReference">
            <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
↳targetClass="PredefinedLocation" version="0"/>
        </pertinentLocation>
        <!-- contained vehicle category -->
        <vehicleType>anyVehicle</vehicleType>
        <!-- travel time information -->
        <travelTime numberOfInputValuesUsed="9" supplierCalculatedDataQuality="1">
            <duration>549</duration>
        </travelTime>
        <!-- historical data from fcd vehicles database -->
        <freeFlowTravelTime numberOfInputValuesUsed="9999"
↳supplierCalculatedDataQuality="1">
            <duration>259</duration>
        </freeFlowTravelTime>
        <!-- free flow vehicle speed information -->
        <freeFlowSpeed numberOfInputValuesUsed="9999" supplierCalculatedDataQuality=
↳"0.8">
            <speed>50</speed>
        </freeFlowSpeed>
        <!-- extension: quality parameters and number of vehicles on the segment NOT
↳present -->
    </basicData>
</elaboratedData>
<!-- travel time element - CARS -->
<elaboratedData>

```

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```

    <basicData xsi:type="TravelTimeData">
      <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
↪ measurementOrCalculationTime>
      <pertinentLocation xsi:type="LocationByReference">
        <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E" ↪
↪ targetClass="PredefinedLocation" version="0"/>
      </pertinentLocation>
      <vehicleType>car</vehicleType>
      <travelTime numberOfInputValuesUsed="5" supplierCalculatedDataQuality="0.850">
        <duration>499</duration>
      </travelTime>
      <freeFlowTravelTime numberOfInputValuesUsed="9999" ↪
↪ supplierCalculatedDataQuality="1">
        <duration>259</duration>
      </freeFlowTravelTime>
      <freeFlowSpeed numberOfInputValuesUsed="9999" supplierCalculatedDataQuality=
↪ "0.8">
        <speed>55</speed>
      </freeFlowSpeed>
    </basicData>
  </elaboratedData>
  <!-- travel time element - LORRIES -->
  <elaboratedData>
    <basicData xsi:type="TravelTimeData">
      <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
↪ measurementOrCalculationTime>
      <pertinentLocation xsi:type="LocationByReference">
        <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E" ↪
↪ targetClass="PredefinedLocation" version="0"/>
      </pertinentLocation>
      <vehicleType>lorry</vehicleType>
      <travelTime numberOfInputValuesUsed="4" supplierCalculatedDataQuality="1">
        <duration>587</duration>
      </travelTime>
      <freeFlowTravelTime numberOfInputValuesUsed="9999" ↪
↪ supplierCalculatedDataQuality="1">
        <duration>279</duration>
      </freeFlowTravelTime>
      <freeFlowSpeed numberOfInputValuesUsed="9999" supplierCalculatedDataQuality=
↪ "0.8">
        <speed>50</speed>
      </freeFlowSpeed>
    </basicData>
  </elaboratedData>
  <!-- information for another segment follows -->
</payloadPublication>
</d2LogicalModel>

```

5.1.1 Root element and the publication envelope

The root element *d2LogicalModel* declares that this is a message using DATEX II.

The *exchange* element identifies the provider of the publication, **ŘSD**.

The *payloadPublication* of type *ElaboratedDataPublication* contains, in this example 9 *elaboratedData* elements with calculated traffic parameters for **ONE** segment.

The *feedType* element specifies the type of publication, by the uri of the schema **x-format:cz-ndic_d2-fcd-v1.1**.

The *publicationTime* element specifies the time of the publication **2022-12-18T00:00:06+01:00**.

The *publicationCreator* element identifies the creator of the publication, **ŘSD**.

The *timeDefault* element indicates the time of calculation of all FCD values in the publication **2022-12-17T23:59:35+01:00**.

5.1.2 General characteristics of the *elaboratedData* elements

A few characteristics are similar/same for all *elaboratedData* elements, they are listed below.

Location reference

The segment location in *predefinedLocationReference* elements refers to data item id **6D476A44-AA44-4947-AA4B-7C4DD79BB67E** with version **0** defined in another publication of type *PredefinedLocationsPublication* which is not described in detail here.

The sample contains FCD information for one physical location (route with identifier **6D476A44-AA44-4947-AA4B-7C4DD79BB67E**).

Temporal validity

The last time of occurrence of vehicle probes on the measured segment is given in the *measurementOrCalculationTime* elements is **2022-12-17T23:57:35+01:00**. This value may be different for each segment and for vehicle type.

Measurement quality

The measurement quality given in the *supplierCalculatedDataQuality* attributes of the calculated values, for all vehicles, is **1**. It means “reliable”. The measurement quality for passenger vehicles is **0.850** and for lorries is **0.750**.

This value should be the same for all actual measurement values on the segment.

Number of vehicles on the segment

The number of vehicle probes on the segment *numberOfInputValuesUsed* at the time of measurement (300 s) is **9**, out of which **4** were lorries and **5** passenger vehicles.

This value should be the same for all the actual measurement values on a given segment.

Source identification

The optional *sourceIdentification* is provided only in first element *elaboratedData* , as an example, and may provide an alternative route identification.

5.1.3 Traffic levels and queues

The first 3 elements *elaboratedData* of type *TrafficStatus* describe calculated traffic status and queues

All 3 elements indicate traffic status (*trafficStatus*) on a four-level scale, as **congested**.

Traffic levels

All 3 elements indicate level of service on a five-level scale, as level 4 = **level4** (equivalent of congested).

Queue length

All 3 elements indicate queues, different per vehicle type.

The aggregate *length* of the queue on the segment at that time was **148 m** for all vehicles and **89 m** for cars and **148 m** for lorries.

The *toPoint* elements specify the same queue end (**0.9** of the segment length).

The *fromPoint* elements specify different start at **0.6** for passenger vehicles and **0.4** for lorries (and all vehicles) in relative coordinates.

5.1.4 Measured speed

The next 3 elements *elaboratedData* of type *TrafficSpeed* describe the computed traffic flow speed.

Traffic speed

All 3 elements indicate current average vehicle speed *averageVehicleSpeed*. This is **38 km/h** for all vehicles and **58 km/h** for passenger vehicles and **18 km/h** for lorries.

5.1.5 Travel time and speed

Last 3 elements *elaboratedData* of type *TravelTimeData* describe the calculated actual and free flow travel times and free flow speed.

Travel time

The *travelTime* element specifies the average current travel time of **549 s** for all vehicles, **499 s** for cars and **587 s** for lorries.

The *freeFlowTravelTime* element gives an average travel time in free flowing traffic of **259 s** for all vehicles, **259 s** for cars and **279 s** for lorries.

Speed

The *freeFlowSpeed* element indicates the average speed in flowing traffic of **50 km/h** for all vehicles, **55 km/h** for cars and **50 km/h** for lorries.

5.2 samples/fcd-all-lorry-car-sample-no-data.xml - no FCD data on 1 segment

The sample contains FCD information for the same location as the previous sample. The difference between the samples is the **unavailability of data to calculate status information**.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsi=
  ↪ "http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://datex2.eu/
  ↪ schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="ElaboratedDataPublication">
    <feedType>x-format:cz-ndic_d2-fcd-v1.1</feedType>
    <publicationTime>2022-12-18T00:00:06+01:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </publicationCreator>
    <periodDefault>300</periodDefault>
    <timeDefault>2022-12-17T23:59:35+01:00</timeDefault>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <!-- information for one segment -->
    <!-- NO traffic status element - no data -->
    <!-- NO speed information element - no data -->
    <!-- travel time element - ALL vehicles -->
    <elaboratedData>
      <basicData xsi:type="TravelTimeData">
        <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
  ↪ measurementOrCalculationTime>
        <pertinentLocation xsi:type="LocationByReference">
          <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
  ↪ ↵ targetClass="PredefinedLocation" version="0"/>
        </pertinentLocation>
        <!-- contained vehicle category -->
        <vehicleType>anyVehicle</vehicleType>
        <!-- historical data from fcd vehicles database -->
        <freeFlowTravelTime numberOfInputValuesUsed="9999"
  ↪ ↵ supplierCalculatedDataQuality="1">
          <duration>259</duration>
        </freeFlowTravelTime>
        <!-- free flow vehicle speed information -->
        <freeFlowSpeed numberOfInputValuesUsed="9999" supplierCalculatedDataQuality=
```

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```

↪ "0.8">
    <speed>50</speed>
  </freeFlowSpeed>
  <!-- extension: quality parameters and number of vehicles on the segment NOT_
↪ present -->
  </basicData>
</elaboratedData>
<!-- travel time element - CARS -->
<elaboratedData>
  <basicData xsi:type="TravelTimeData">
    <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
↪ measurementOrCalculationTime>
    <pertinentLocation xsi:type="LocationByReference">
      <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
↪ targetClass="PredefinedLocation" version="0"/>
    </pertinentLocation>
    <vehicleType>car</vehicleType>
    <freeFlowTravelTime numberOfInputValuesUsed="9999"
↪ supplierCalculatedDataQuality="1">
      <duration>259</duration>
    </freeFlowTravelTime>
    <freeFlowSpeed numberOfInputValuesUsed="9999" supplierCalculatedDataQuality=
↪ "0.8">
      <speed>55</speed>
    </freeFlowSpeed>
  </basicData>
</elaboratedData>
<!-- travel time element - LORRIES -->
<elaboratedData>
  <basicData xsi:type="TravelTimeData">
    <measurementOrCalculationTime>2022-12-17T23:57:35+01:00</
↪ measurementOrCalculationTime>
    <pertinentLocation xsi:type="LocationByReference">
      <predefinedLocationReference id="6D476A44-AA44-4947-AA4B-7C4DD79BB67E"
↪ targetClass="PredefinedLocation" version="0"/>
    </pertinentLocation>
    <vehicleType>lorry</vehicleType>
    <freeFlowTravelTime numberOfInputValuesUsed="9999"
↪ supplierCalculatedDataQuality="1">
      <duration>279</duration>
    </freeFlowTravelTime>
    <freeFlowSpeed numberOfInputValuesUsed="9999" supplierCalculatedDataQuality=
↪ "0.8">
      <speed>50</speed>
    </freeFlowSpeed>
  </basicData>
</elaboratedData>
  <!-- information for another segment follows -->
</payloadPublication>
</d2LogicalModel>

```

In the sections below, we only list the differences from the previous sample.

5.2.1 General characteristics of the *elaboratedData* elements

A few characteristics are similar/same for all *elaboratedData* elements, they are listed below.

Temporal validity

The last time of occurrence of vehicle probes on the measured segment is given in the *measurementOrCalculationTime* elements is **2022-12-17T22:57:35+01:00**, about an hour ago.

Measurement quality

Irrelevant, no data to calculate.

Number of vehicles on the segment

Irrelevant, no data to calculate.

5.2.2 Traffic levels and queues

Irrelevant, no data to calculate.

5.2.3 Measured speed

Irrelevant, no data to calculate.

5.2.4 Travel time and speed

The 3 elements *elaboratedData* of type *TravelTimeData* contains only information about the free flowing traffic flow.

Travel time

The *freeFlowTravelTime* element gives an average travel time in free flowing traffic of **259 s** for all vehicles, **259 s** for cars and **279 s** for lorries.

Speed

The *freeFlowSpeed* element indicates an average speed in flowing traffic of **50 km/h** for all vehicles, **55 km/h** for cars and **50 km/h** for lorries.

APPENDIX B: W3C XML SCHEMA

XML structure of a message is defined by following W3C XML schema:

6.1 *schema.xsd*

```

1 <?xml version="1.0" encoding="utf-8" standalone="no"?>
2 <xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified"
  ↳ xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace=
  ↳ "http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:complexType name="_ExtensionType">
4     <xs:sequence>
5       <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
6     </xs:sequence>
7   </xs:complexType>
8   <xs:complexType name="_PredefinedLocationVersionedReference">
9     <xs:complexContent>
10      <xs:extension base="D2LogicalModel:VersionedReference">
11        <xs:attribute name="targetClass" use="required" fixed="PredefinedLocation" />
12      </xs:extension>
13    </xs:complexContent>
14  </xs:complexType>
15  <xs:complexType name="_TrafficStatusExtensionType">
16    <xs:sequence>
17      <xs:element name="ndicFcdExtension" type="D2LogicalModel:NdicFcdExtension"
  ↳ minOccurs="0" />
18      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
19    </xs:sequence>
20  </xs:complexType>
21  <xs:complexType name="_TravelTimeDataExtensionType">
22    <xs:sequence>
23      <xs:element name="qualityParameters" type="D2LogicalModel:QualityParameters"
  ↳ minOccurs="0" />
24      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
25    </xs:sequence>
26  </xs:complexType>
27  <xs:simpleType name="AreaOfInterestEnum">
28    <xs:restriction base="xs:string">
29      <xs:enumeration value="continentWide" />
30      <xs:enumeration value="national" />
31      <xs:enumeration value="neighbouringCountries" />
32      <xs:enumeration value="notSpecified" />

```

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```

33     <xs:enumeration value="regional" />
34   </xs:restriction>
35 </xs:simpleType>
36 <xs:complexType name="BasicData" abstract="true">
37   <xs:sequence>
38     <xs:element name="measurementOrCalculationTime" type="D2LogicalModel:DateTime"
↪ minOccurs="0" maxOccurs="1" />
39     <xs:element name="pertinentLocation" type="D2LogicalModel:GroupOfLocations"
↪ minOccurs="0" />
40     <xs:element name="basicDataExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
41   </xs:sequence>
42 </xs:complexType>
43 <xs:simpleType name="Boolean">
44   <xs:restriction base="xs:boolean" />
45 </xs:simpleType>
46 <xs:simpleType name="ComputationMethodEnum">
47   <xs:restriction base="xs:string">
48     <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples" /
↪ >
49     <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
50     <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
51     <xs:enumeration value="medianOfSamplesInATimePeriod" />
52     <xs:enumeration value="movingAverageOfSamples" />
53   </xs:restriction>
54 </xs:simpleType>
55 <xs:simpleType name="ConfidentialityValueEnum">
56   <xs:restriction base="xs:string">
57     <xs:enumeration value="internalUse" />
58     <xs:enumeration value="noRestriction" />
59     <xs:enumeration value="restrictedToAuthorities" />
60     <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
61     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
62     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
63   </xs:restriction>
64 </xs:simpleType>
65 <xs:simpleType name="CountryEnum">
66   <xs:restriction base="xs:string">
67     <xs:enumeration value="at" />
68     <xs:enumeration value="be" />
69     <xs:enumeration value="bg" />
70     <xs:enumeration value="ch" />
71     <xs:enumeration value="cs" />
72     <xs:enumeration value="cy" />
73     <xs:enumeration value="cz" />
74     <xs:enumeration value="de" />
75     <xs:enumeration value="dk" />
76     <xs:enumeration value="ee" />
77     <xs:enumeration value="es" />
78     <xs:enumeration value="fi" />
79     <xs:enumeration value="fo" />
80     <xs:enumeration value="fr" />
81     <xs:enumeration value="gb" />
82     <xs:enumeration value="gg" />
83     <xs:enumeration value="gi" />
84     <xs:enumeration value="gr" />

```

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```

85     <xs:enumeration value="hr" />
86     <xs:enumeration value="hu" />
87     <xs:enumeration value="ie" />
88     <xs:enumeration value="im" />
89     <xs:enumeration value="is" />
90     <xs:enumeration value="it" />
91     <xs:enumeration value="je" />
92     <xs:enumeration value="li" />
93     <xs:enumeration value="lt" />
94     <xs:enumeration value="lu" />
95     <xs:enumeration value="lv" />
96     <xs:enumeration value="ma" />
97     <xs:enumeration value="mc" />
98     <xs:enumeration value="mk" />
99     <xs:enumeration value="mt" />
100    <xs:enumeration value="nl" />
101    <xs:enumeration value="no" />
102    <xs:enumeration value="pl" />
103    <xs:enumeration value="pt" />
104    <xs:enumeration value="ro" />
105    <xs:enumeration value="se" />
106    <xs:enumeration value="si" />
107    <xs:enumeration value="sk" />
108    <xs:enumeration value="sm" />
109    <xs:enumeration value="tr" />
110    <xs:enumeration value="va" />
111    <xs:enumeration value="other" />
112  </xs:restriction>
113 </xs:simpleType>
114 <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel" />
115 <xs:complexType name="D2LogicalModel">
116   <xs:sequence>
117     <xs:element name="exchange" type="D2LogicalModel:Exchange" />
118     <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication"
119     ↪ minOccurs="0" />
120     <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType"
121     ↪ minOccurs="0" />
122   </xs:sequence>
123   <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
124 </xs:complexType>
125 <xs:complexType name="DataValue" abstract="true">
126   <xs:sequence>
127     <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0"
128     ↪ maxOccurs="1" />
129     <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString"
130     ↪ minOccurs="0" maxOccurs="1" />
131     <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType"
132     ↪ minOccurs="0" />
133   </xs:sequence>
134   <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" />
135   <xs:attribute name="computationalMethod" type=
136   ↪ "D2LogicalModel:ComputationMethodEnum" use="optional" />
137   <xs:attribute name="numberOfIncompleteInputs" type=
138   ↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
139   <xs:attribute name="numberOfInputValuesUsed" type=
140   ↪ "D2LogicalModel:NonNegativeInteger" use="optional" />

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```

133     <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional" />
134     <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional" />
135     </xs:complexType>
136     <xs:simpleType name="DateTime">
137       <xs:restriction base="xs:dateTime" />
138     </xs:simpleType>
139     <xs:complexType name="DurationValue">
140       <xs:complexContent>
141         <xs:extension base="D2LogicalModel:DataValue">
142           <xs:sequence>
143             <xs:element name="duration" type="D2LogicalModel:Seconds" minOccurs="1"
144             ↪ maxOccurs="1" />
145             <xs:element name="durationValueExtension" type="D2LogicalModel:_
146             ↪ ExtensionType" minOccurs="0" />
147           </xs:sequence>
148         </xs:extension>
149       </xs:complexContent>
150     </xs:complexType>
151     <xs:complexType name="ElaboratedData">
152       <xs:sequence>
153         <xs:element name="source" type="D2LogicalModel:Source" minOccurs="0" />
154         <xs:element name="elaboratedDataFault" type="D2LogicalModel:ElaboratedDataFault
155         ↪ " minOccurs="0" maxOccurs="unbounded" />
156         <xs:element name="basicData" type="D2LogicalModel:BasicData" minOccurs="0" />
157         <xs:element name="elaboratedDataExtension" type="D2LogicalModel:_ExtensionType"
158         ↪ minOccurs="0" />
159       </xs:sequence>
160     </xs:complexType>
161     <xs:complexType name="ElaboratedDataFault">
162       <xs:complexContent>
163         <xs:extension base="D2LogicalModel:Fault">
164           <xs:sequence>
165             <xs:element name="elaboratedDataFault" type=
166             ↪ "D2LogicalModel:ElaboratedDataFaultEnum" minOccurs="1" maxOccurs="1" />
167             <xs:element name="elaboratedDataFaultExtension" type="D2LogicalModel:_
168             ↪ ExtensionType" minOccurs="0" />
169           </xs:sequence>
170         </xs:extension>
171       </xs:complexContent>
172     </xs:complexType>
173     <xs:simpleType name="ElaboratedDataFaultEnum">
174       <xs:restriction base="xs:string">
175         <xs:enumeration value="intermittentDataValues" />
176         <xs:enumeration value="noDataValuesAvailable" />
177         <xs:enumeration value="spuriousUnreliableDataValues" />
178         <xs:enumeration value="unspecifiedOrUnknownFault" />
179         <xs:enumeration value="other" />
180       </xs:restriction>
181     </xs:simpleType>
182     <xs:complexType name="ElaboratedDataPublication">
183       <xs:complexContent>
184         <xs:extension base="D2LogicalModel:PayloadPublication">
185           <xs:sequence>

```

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```

181     <xs:element name="periodDefault" type="D2LogicalModel:Seconds" minOccurs="0
↪ " maxOccurs="1" />
182     <xs:element name="timeDefault" type="D2LogicalModel:DateTime" minOccurs="0"
↪ maxOccurs="1" />
183     <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation
↪ " />
184     <xs:element name="elaboratedData" type="D2LogicalModel:ElaboratedData"
↪ maxOccurs="unbounded" />
185     <xs:element name="elaboratedDataPublicationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
186     </xs:sequence>
187     </xs:extension>
188     </xs:complexContent>
189   </xs:complexType>
190   <xs:complexType name="Exchange">
191     <xs:sequence>
192       <xs:element name="supplierIdentification" type=
↪ "D2LogicalModel:InternationalIdentifier" />
193       <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
194     </xs:sequence>
195   </xs:complexType>
196   <xs:complexType name="Fault">
197     <xs:sequence>
198       <xs:element name="faultIdentifier" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
199       <xs:element name="faultDescription" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
200       <xs:element name="faultCreationTime" type="D2LogicalModel:DateTime" minOccurs="0
↪ " maxOccurs="1" />
201       <xs:element name="faultLastUpdateTime" type="D2LogicalModel:DateTime" minOccurs=
↪ "1" maxOccurs="1" />
202       <xs:element name="faultSeverity" type="D2LogicalModel:FaultSeverityEnum"
↪ minOccurs="0" maxOccurs="1" />
203       <xs:element name="faultExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
204     </xs:sequence>
205   </xs:complexType>
206   <xs:simpleType name="FaultSeverityEnum">
207     <xs:restriction base="xs:string">
208       <xs:enumeration value="low" />
209       <xs:enumeration value="medium" />
210       <xs:enumeration value="high" />
211       <xs:enumeration value="unknown" />
212     </xs:restriction>
213   </xs:simpleType>
214   <xs:simpleType name="Float">
215     <xs:restriction base="xs:float" />
216   </xs:simpleType>
217   <xs:complexType name="GroupOfLocations" abstract="true">
218     <xs:sequence>
219       <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType
↪ " minOccurs="0" />
220     </xs:sequence>
221   </xs:complexType>
222   <xs:complexType name="HeaderInformation">

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223 <xs:sequence>
224   <xs:element name="areaOfInterest" type="D2LogicalModel:AreaOfInterestEnum"
↪ minOccurs="0" maxOccurs="1" />
225   <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum
↪ " minOccurs="1" maxOccurs="1" />
226   <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum
↪ " minOccurs="1" maxOccurs="1" />
227   <xs:element name="headerInformationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
228 </xs:sequence>
229 </xs:complexType>
230 <xs:simpleType name="InformationStatusEnum">
231   <xs:restriction base="xs:string">
232     <xs:enumeration value="real" />
233     <xs:enumeration value="securityExercise" />
234     <xs:enumeration value="technicalExercise" />
235     <xs:enumeration value="test" />
236   </xs:restriction>
237 </xs:simpleType>
238 <xs:complexType name="InternationalIdentifier">
239   <xs:sequence>
240     <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"
↪ maxOccurs="1" />
241     <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1"
↪ " maxOccurs="1" />
242     <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
243   </xs:sequence>
244 </xs:complexType>
245 <xs:simpleType name="KilometresPerHour">
246   <xs:restriction base="D2LogicalModel:Float" />
247 </xs:simpleType>
248 <xs:simpleType name="Language">
249   <xs:restriction base="xs:language" />
250 </xs:simpleType>
251 <xs:complexType name="Location" abstract="true">
252   <xs:complexContent>
253     <xs:extension base="D2LogicalModel:GroupOfLocations">
254       <xs:sequence>
255         <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
256       </xs:sequence>
257     </xs:extension>
258   </xs:complexContent>
259 </xs:complexType>
260 <xs:complexType name="LocationByReference">
261   <xs:complexContent>
262     <xs:extension base="D2LogicalModel:Location">
263       <xs:sequence>
264         <xs:element name="predefinedLocationReference" type="D2LogicalModel:_
↪ PredefinedLocationVersionedReference" minOccurs="1" maxOccurs="1" />
265         <xs:element name="locationByReferenceExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
266       </xs:sequence>
267     </xs:extension>
268   </xs:complexContent>

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269 </xs:complexType>
270 <xs:complexType name="MeasurementEquipmentFault">
271   <xs:complexContent>
272     <xs:extension base="D2LogicalModel:Fault">
273       <xs:sequence>
274         <xs:element name="measurementEquipmentFault" type=
275 ↪ "D2LogicalModel:MeasurementEquipmentFaultEnum" minOccurs="1" maxOccurs="1" />
276         <xs:element name="measurementEquipmentFaultExtension" type="D2LogicalModel:_
277 ↪ ExtensionType" minOccurs="0" />
278       </xs:sequence>
279     </xs:extension>
280   </xs:complexContent>
281 </xs:complexType>
282 <xs:simpleType name="MeasurementEquipmentFaultEnum">
283   <xs:restriction base="xs:string">
284     <xs:enumeration value="intermittentDataValues" />
285     <xs:enumeration value="noDataValuesAvailable" />
286     <xs:enumeration value="spuriousUnreliableDataValues" />
287     <xs:enumeration value="unspecifiedOrUnknownFault" />
288     <xs:enumeration value="other" />
289   </xs:restriction>
290 </xs:simpleType>
291 <xs:simpleType name="MetresAsNonNegativeInteger">
292   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
293 </xs:simpleType>
294 <xs:complexType name="MultilingualString">
295   <xs:sequence>
296     <xs:element name="values">
297       <xs:complexType>
298         <xs:sequence>
299           <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"
300 ↪ maxOccurs="unbounded" />
301         </xs:sequence>
302       </xs:complexType>
303     </xs:element>
304   </xs:sequence>
305 </xs:complexType>
306 <xs:complexType name="MultilingualStringValue">
307   <xs:simpleContent>
308     <xs:extension base="D2LogicalModel:MultilingualStringValueType">
309       <xs:attribute name="lang" type="xs:language" />
310     </xs:extension>
311   </xs:simpleContent>
312 </xs:complexType>
313 <xs:simpleType name="MultilingualStringValueType">
314   <xs:restriction base="xs:string">
315     <xs:maxLength value="1024" />
316   </xs:restriction>
317 </xs:simpleType>
318 <xs:complexType name="NdicFcdExtension">
319   <xs:sequence>
320     <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs=
321 ↪ "0" maxOccurs="3" />
322     <xs:element name="trafficLevel" type="D2LogicalModel:TrafficLevel" minOccurs="0
323 ↪ " />
324     <xs:element name="queueInformation" type="D2LogicalModel:QueueInformation"

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320     </xs:sequence>
321 </xs:complexType>
322 <xs:simpleType name="NonNegativeInteger">
323   <xs:restriction base="xs:nonNegativeInteger" />
324 </xs:simpleType>
325 <xs:complexType name="NumberOfFcdVehicles">
326   <xs:sequence>
327     <xs:element name="vehicleGroup" type="D2LogicalModel:VehicleGroup" maxOccurs=
328     ↪ "unbounded" />
329     <xs:element name="numberOfFcdVehiclesExtension" type="D2LogicalModel:_
330     ↪ ExtensionType" minOccurs="0" />
331   </xs:sequence>
332 </xs:complexType>
333 <xs:complexType name="PayloadPublication" abstract="true">
334   <xs:sequence>
335     <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0" ↪
336     ↪ maxOccurs="1" />
337     <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1" ↪
338     ↪ maxOccurs="1" />
339     <xs:element name="publicationCreator" type=
340     ↪ "D2LogicalModel:InternationalIdentifier" />
341     <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
342     ↪ ExtensionType" minOccurs="0" />
343   </xs:sequence>
344   <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
345 </xs:complexType>
346 <xs:simpleType name="Percentage">
347   <xs:restriction base="D2LogicalModel:Float" />
348 </xs:simpleType>
349 <xs:complexType name="PercentageValue">
350   <xs:complexContent>
351     <xs:extension base="D2LogicalModel:DataValue">
352       <xs:sequence>
353         <xs:element name="percentage" type="D2LogicalModel:Percentage" minOccurs="1"
354         ↪ " maxOccurs="1" />
355         <xs:element name="percentageValueExtension" type="D2LogicalModel:_
356         ↪ ExtensionType" minOccurs="0" />
357       </xs:sequence>
358     </xs:extension>
359   </xs:complexContent>
360 </xs:complexType>
361 <xs:complexType name="QualityParameters">
362   <xs:sequence>
363     <xs:element name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage"
364     ↪ " minOccurs="1" maxOccurs="1" />
365     <xs:element name="numberOfFcdVehicles" type="D2LogicalModel:NumberOfFcdVehicles"
366     ↪ " minOccurs="0" />
367   </xs:sequence>
368 </xs:complexType>
369 <xs:complexType name="Queue">
370   <xs:complexContent>
371     <xs:extension base="D2LogicalModel:DataValue">
372       <xs:sequence>
373         <xs:element name="queueLength" type=
374         ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="1" maxOccurs="1" />

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364     <xs:element name="fromPoint" type="D2LogicalModel:Percentage" minOccurs="0"
↪ maxOccurs="1" />
365     <xs:element name="toPoint" type="D2LogicalModel:Percentage" minOccurs="0"
↪ maxOccurs="1" />
366     <xs:element name="queueExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
367     </xs:sequence>
368   </xs:extension>
369 </xs:complexContent>
370 </xs:complexType>
371 <xs:complexType name="QueueInformation">
372   <xs:sequence>
373     <xs:element name="queue" type="D2LogicalModel:Queue" maxOccurs="unbounded" />
374     <xs:element name="queueInformationExtension" type="D2LogicalModel:_ExtensionType"
↪ " minOccurs="0" />
375   </xs:sequence>
376 </xs:complexType>
377 <xs:simpleType name="Seconds">
378   <xs:restriction base="D2LogicalModel:Float" />
379 </xs:simpleType>
380 <xs:complexType name="Source">
381   <xs:sequence>
382     <xs:element name="sourceIdentification" type="D2LogicalModel:String" minOccurs=
↪ "0" maxOccurs="1" />
383     <xs:element name="sourceExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
384   </xs:sequence>
385 </xs:complexType>
386 <xs:complexType name="SpeedPercentile">
387   <xs:sequence>
388     <xs:element name="vehiclePercentage" type="D2LogicalModel:PercentageValue" />
389     <xs:element name="speedPercentile" type="D2LogicalModel:SpeedValue" />
390     <xs:element name="speedPercentileExtension" type="D2LogicalModel:_ExtensionType"
↪ " minOccurs="0" />
391   </xs:sequence>
392 </xs:complexType>
393 <xs:complexType name="SpeedValue">
394   <xs:complexContent>
395     <xs:extension base="D2LogicalModel:DataValue">
396       <xs:sequence>
397         <xs:element name="speed" type="D2LogicalModel:KilometresPerHour" minOccurs=
↪ "1" maxOccurs="1" />
398         <xs:element name="speedValueExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
399       </xs:sequence>
400     </xs:extension>
401   </xs:complexContent>
402 </xs:complexType>
403 <xs:simpleType name="String">
404   <xs:restriction base="xs:string">
405     <xs:maxLength value="1024" />
406   </xs:restriction>
407 </xs:simpleType>
408 <xs:complexType name="TrafficData" abstract="true">
409   <xs:complexContent>
410     <xs:extension base="D2LogicalModel:BasicData">

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411     <xs:sequence>
412       <xs:element name="forVehiclesWithCharacteristicsOf" type=
↪ "D2LogicalModel:VehicleCharacteristics" minOccurs="0" />
413       <xs:element name="trafficDataExtension" type="D2LogicalModel:_ExtensionType"
↪ " minOccurs="0" />
414     </xs:sequence>
415   </xs:extension>
416 </xs:complexContent>
417 </xs:complexType>
418 <xs:complexType name="TrafficLevel">
419   <xs:complexContent>
420     <xs:extension base="D2LogicalModel:DataValue">
421       <xs:sequence>
422         <xs:element name="trafficLevelValue" type="D2LogicalModel:TrafficLevelEnum"
↪ minOccurs="1" maxOccurs="1" />
423         <xs:element name="trafficLevelExtension" type="D2LogicalModel:_ExtensionType"
↪ " minOccurs="0" />
424       </xs:sequence>
425     </xs:extension>
426   </xs:complexContent>
427 </xs:complexType>
428 <xs:simpleType name="TrafficLevelEnum">
429   <xs:restriction base="xs:string" />
430 </xs:simpleType>
431 <xs:complexType name="TrafficSpeed">
432   <xs:complexContent>
433     <xs:extension base="D2LogicalModel:TrafficData">
434       <xs:sequence>
435         <xs:element name="averageVehicleSpeed" type="D2LogicalModel:SpeedValue"
↪ minOccurs="0" />
436         <xs:element name="speedPercentile" type="D2LogicalModel:SpeedPercentile"
↪ minOccurs="0" />
437         <xs:element name="trafficSpeedExtension" type="D2LogicalModel:_ExtensionType"
↪ " minOccurs="0" />
438       </xs:sequence>
439     </xs:extension>
440   </xs:complexContent>
441 </xs:complexType>
442 <xs:complexType name="TrafficStatus">
443   <xs:complexContent>
444     <xs:extension base="D2LogicalModel:BasicData">
445       <xs:sequence>
446         <xs:element name="trafficStatus" type="D2LogicalModel:TrafficStatusValue"
↪ minOccurs="0" />
447         <xs:element name="trafficStatusExtension" type="D2LogicalModel:_
↪ TrafficStatusExtensionType" minOccurs="0" />
448       </xs:sequence>
449     </xs:extension>
450   </xs:complexContent>
451 </xs:complexType>
452 <xs:simpleType name="TrafficStatusEnum">
453   <xs:restriction base="xs:string">
454     <xs:enumeration value="impossible" />
455     <xs:enumeration value="congested" />
456     <xs:enumeration value="heavy" />
457     <xs:enumeration value="freeFlow" />

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458     <xs:enumeration value="unknown" />
459   </xs:restriction>
460 </xs:simpleType>
461 <xs:complexType name="TrafficStatusValue">
462   <xs:complexContent>
463     <xs:extension base="D2LogicalModel:DataValue">
464       <xs:sequence>
465         <xs:element name="trafficStatusValue" type="D2LogicalModel:TrafficStatusEnum
466 ↪ " minOccurs="1" maxOccurs="1" />
467         <xs:element name="trafficStatusValueExtension" type="D2LogicalModel:_
468 ↪ ExtensionType" minOccurs="0" />
469       </xs:sequence>
470     </xs:extension>
471   </xs:complexContent>
472 </xs:complexType>
473 <xs:complexType name="TravelTimeData">
474   <xs:complexContent>
475     <xs:extension base="D2LogicalModel:BasicData">
476       <xs:sequence>
477         <xs:element name="travelTimeType" type="D2LogicalModel:TravelTimeTypeEnum"
478 ↪ minOccurs="0" maxOccurs="1" />
479         <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum"
480 ↪ minOccurs="0" maxOccurs="unbounded" />
481         <xs:element name="travelTime" type="D2LogicalModel:DurationValue" minOccurs=
482 ↪ "0" />
483         <xs:element name="freeFlowTravelTime" type="D2LogicalModel:DurationValue"
484 ↪ minOccurs="0" />
485         <xs:element name="normallyExpectedTravelTime" type=
486 ↪ "D2LogicalModel:DurationValue" minOccurs="0" />
487         <xs:element name="freeFlowSpeed" type="D2LogicalModel:SpeedValue" minOccurs=
488 ↪ "0" />
489         <xs:element name="travelTimeDataExtension" type="D2LogicalModel:_
490 ↪ TravelTimeDataExtensionType" minOccurs="0" />
491       </xs:sequence>
492     </xs:extension>
493   </xs:complexContent>
494 </xs:complexType>
495 <xs:simpleType name="TravelTimeTypeEnum">
496   <xs:restriction base="xs:string">
497     <xs:enumeration value="best" />
498     <xs:enumeration value="estimated" />
499     <xs:enumeration value="instantaneous" />
500     <xs:enumeration value="reconstituted" />
501   </xs:restriction>
502 </xs:simpleType>
503 <xs:complexType name="VehicleCharacteristics">
504   <xs:sequence>
505     <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs=
506 ↪ "0" maxOccurs="unbounded" />
507     <xs:element name="vehicleCharacteristicsExtension" type="D2LogicalModel:_
508 ↪ ExtensionType" minOccurs="0" />
509   </xs:sequence>
510 </xs:complexType>
511 <xs:complexType name="VehicleGroup">
512   <xs:sequence>
513     <xs:element name="count" type="D2LogicalModel:NonNegativeInteger" minOccurs="1"

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503   <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs=
504   <xs:element name="vehicleGroupExtension" type="D2LogicalModel:_ExtensionType"
505   </xs:sequence>
506 </xs:complexType>
507 <xs:simpleType name="VehicleTypeEnum">
508   <xs:restriction base="xs:string">
509     <xs:enumeration value="anyVehicle" />
510     <xs:enumeration value="car" />
511     <xs:enumeration value="lorry" />
512   </xs:restriction>
513 </xs:simpleType>
514 <xs:complexType name="VersionedReference">
515   <xs:attribute name="id" type="xs:string" use="required" />
516   <xs:attribute name="version" type="xs:string" use="required" />
517 </xs:complexType>
518 <xs:complexType name="VmsFault">
519   <xs:complexContent>
520     <xs:extension base="D2LogicalModel:Fault">
521       <xs:sequence>
522         <xs:element name="vmsFault" type="D2LogicalModel:VmsFaultEnum" minOccurs="1
523         <xs:element name="vmsFaultExtension" type="D2LogicalModel:_ExtensionType"
524         </xs:sequence>
525       </xs:extension>
526     </xs:complexContent>
527 </xs:complexType>
528 <xs:simpleType name="VmsFaultEnum">
529   <xs:restriction base="xs:string">
530     <xs:enumeration value="communicationsFailure" />
531     <xs:enumeration value="incorrectMessageDisplayed" />
532     <xs:enumeration value="incorrectPictogramDisplayed" />
533     <xs:enumeration value="outOfService" />
534     <xs:enumeration value="powerFailure" />
535     <xs:enumeration value="unableToClearDown" />
536     <xs:enumeration value="unknown" />
537     <xs:enumeration value="other" />
538   </xs:restriction>
539 </xs:simpleType>
540 <xs:complexType name="VmsUnitFault">
541   <xs:complexContent>
542     <xs:extension base="D2LogicalModel:Fault">
543       <xs:sequence>
544         <xs:element name="vmsUnitFault" type="D2LogicalModel:VmsFaultEnum"
545         <xs:element name="vmsUnitFaultExtension" type="D2LogicalModel:_ExtensionType
546         </xs:sequence>
547       </xs:extension>
548     </xs:complexContent>
549 </xs:complexType>
550 </xs:schema>

```

CHANGELOG

version 1.1.0

- format updated to allow for vehicle type partial feed (i.e. for lorries and vehicles)

version 1.0.0

- Initial format and documentation

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Symbols

(DATEX II) extension, [3](#)
(DATEX II) profile, [2](#)
(W3C) XML schema, [3](#)

B

B level extension, [3](#)

C

C level extension, [3](#)

D

DATEX II model, [2](#)

F

format, [2](#)

N

NDIC, [2](#)

P

protocol, [2](#)
publication, [3](#)

U

uri, [3](#)

X

XSD, [3](#)